



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO ARD AGENCY

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

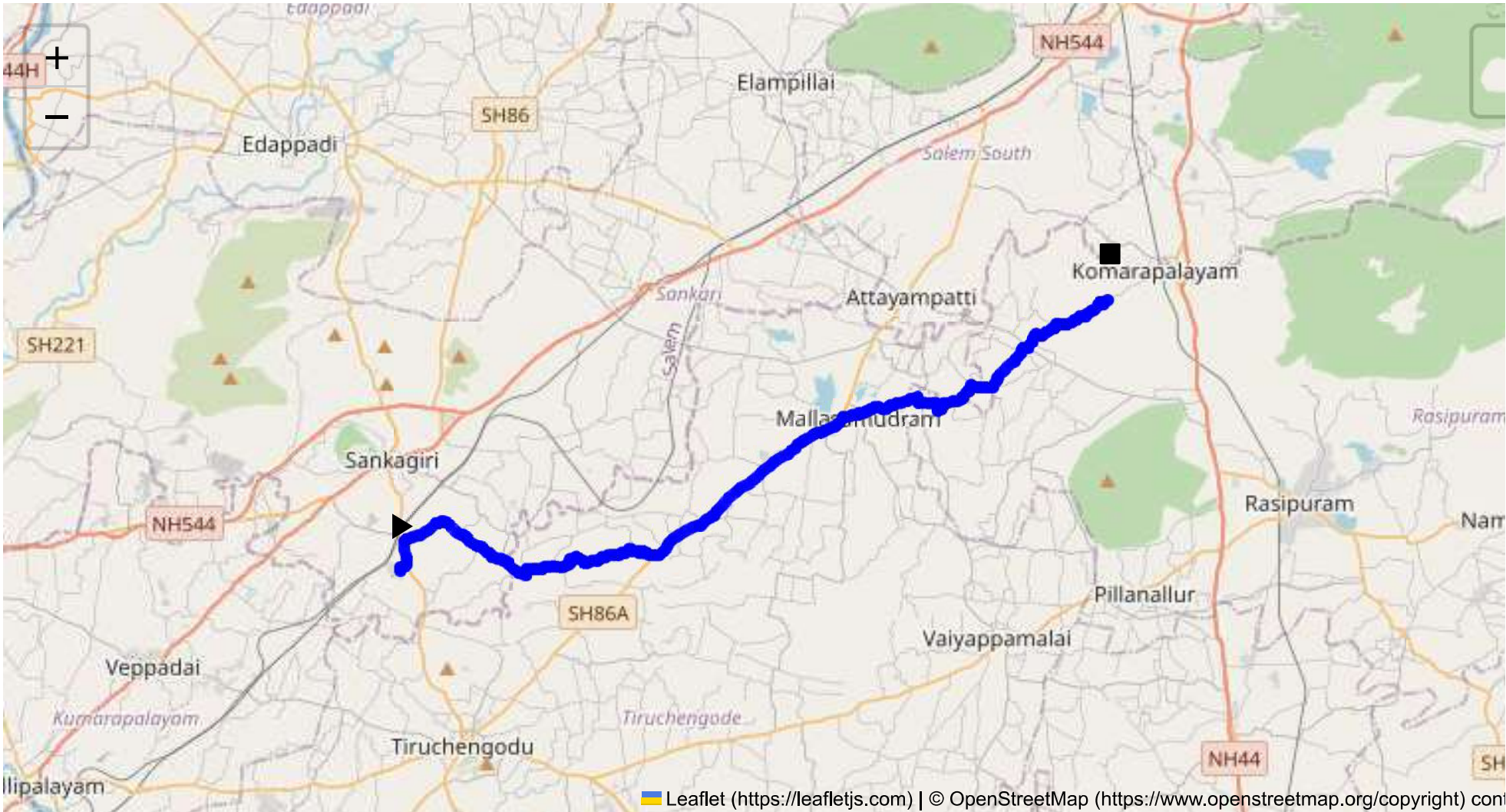
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 34.59 km
Estimated Duration: 1.0 hours
Adjusted Duration (Heavy Vehicle): 1.2 hours
Start: (11.4381, 77.8734)
End: (11.5308273, 78.1189141)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from Sangagiri to Ponparappipatti extends over approximately 34.59 kilometers and takes nearly an hour for heavy vehicles. It primarily utilizes state and local roads, passing through rural and semi-urban areas. The roadway may comprise narrow two-lane roads typical of Tamil Nadu’s landscape.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu’s climate is generally tropical, with significant seasonal variation:

- **Monsoon Season (June to September):** Heavy rains could lead to flooding, slippery roads, and reduced visibility. Monitor for landslides in hilly areas, especially near riverbanks.
- **Summer (March to May):** High temperatures can cause road surface distress and increase the risk of tire blowouts.
- **Winter (December to February):** Mild and generally dry, but early morning fog can affect visibility.

3. Traffic Patterns

- **Peak Hours:** Morning (7-9 AM) and evening (5-8 PM) coincide with local commuters.
- **Congestion Zones:** Urban settlements along the NH 544 and near major intersections might see congestion; checkpoints for checking hazardous materials could also slow down traffic.

4. Assessment of Road Quality and Infrastructure

- **General Condition:** Roads may show uneven surfaces with occasional potholes, especially after monsoon seasons. Regular maintenance is vital but not guaranteed.
- **Infrastructure:** Adequate signage exists for guidance, but lighting may be poor in rural stretches at night.

5. Suggestions for Alternative Routes for Emergencies

- **Alternative Route:** Consider NH 44 for detours, but verify it can handle truck traffic and hazardous materials, as some areas may have restrictions.

6. Summary of Local Regulations Affecting Hazardous Material Transport

- **Timing Restrictions:** Movement might be restricted during peak times and late at night in populated areas.
- **Route Permits:** Special permits may be needed, and routes for hazardous materials are often predefined for safety.

7. Overview of Historical Incidents

While specific historical incident data is sparse, local reports suggest occasional accidents involving heavy vehicles during adverse weather. Emphasize caution during monsoon seasons and on curvy, narrow roads.

8. Environmental Considerations and Sensitive Areas

- **Sensitive Areas:** Rivers and natural reserves nearby; Avoid spills to prevent environmental contamination.
- **Local Flora and Fauna:** Respect restricted zones and be aware of animal crossings.

9. Analysis of Communication Coverage

- **Potential Dead Zones:** Rural areas between towns might experience mobile network dead zones. Ensure communication devices are checked and have alternative communication methods.

10. Estimated Emergency Response Times

- **Urban Areas:** Approximately 20-30 minutes, given the proximity to police stations and fire services.
- **Rural Areas:** Can extend to 40-60 minutes due to terrain and distance.

12. Overall Summary of Risk Assessment

The route is moderately risky due to variable weather, challenging road conditions, and potential traffic congestions. Effective risk management requires awareness of weather forecasts, vehicle checks, and compliance with local regulations. Emergency preparedness is crucial, with alternative routes and communication plans in place. Ensure drivers are trained on local regulations and emergency responses. Regular updates and local coordination will enhance safety on this route.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	Medium	11.43822, 77.87348	30 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44029, 77.87544	15 KM/Hr
3	Blind Spot	Blind Spot	11.44861, 77.87429	10 KM/Hr
4	Turn	Medium	11.45478, 77.88925	30 KM/Hr
5	Turn	High	11.43675, 77.91693	15 KM/Hr
6	Turn	High	11.43684, 77.91706	15 KM/Hr
7	Turn	High	11.43926, 77.91705	15 KM/Hr
8	Turn	Medium	11.43930, 77.92282	30 KM/Hr
9	Turn	Medium	11.43967, 77.92345	30 KM/Hr
10	Turn	Medium	11.43979, 77.93009	30 KM/Hr
11	Turn	Medium	11.44030, 77.93066	30 KM/Hr
12	Turn	Medium	11.44067, 77.93203	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
13	Turn	High	11.44088, 77.93223	15 KM/Hr
14	Turn	Medium	11.44127, 77.93218	30 KM/Hr
15	Turn	Medium	11.44163, 77.93244	30 KM/Hr
16	Turn	High	11.44243, 77.93244	15 KM/Hr
17	Turn	Medium	11.44285, 77.93440	30 KM/Hr
18	Turn	Medium	11.44093, 77.93821	30 KM/Hr
19	Turn	Medium	11.44550, 77.95329	30 KM/Hr
20	Turn	Medium	11.44369, 77.96036	30 KM/Hr
21	Turn	Medium	11.44391, 77.96277	30 KM/Hr
22	Turn	High	11.49063, 78.02680	15 KM/Hr
23	Turn	High	11.49135, 78.03252	15 KM/Hr
24	Turn	High	11.49183, 78.03257	15 KM/Hr
25	Turn	Medium	11.49315, 78.03664	30 KM/Hr
26	Turn	Medium	11.49355, 78.03703	30 KM/Hr
27	Turn	Medium	11.49374, 78.04232	30 KM/Hr
28	Turn	Medium	11.49645, 78.04990	30 KM/Hr
29	Turn	High	11.49676, 78.05010	15 KM/Hr
30	Blind Spot	Blind Spot	11.49723, 78.05308	10 KM/Hr
31	Turn	High	11.49553, 78.05338	15 KM/Hr
32	Turn	Medium	11.49547, 78.05442	30 KM/Hr
33	Turn	High	11.49518, 78.05498	15 KM/Hr
34	Turn	Medium	11.49544, 78.05538	30 KM/Hr
35	Turn	Medium	11.49525, 78.05686	30 KM/Hr
36	Turn	Medium	11.49546, 78.05719	30 KM/Hr
37	Turn	High	11.49548, 78.05904	15 KM/Hr
38	Turn	Medium	11.49477, 78.05905	30 KM/Hr
39	Turn	High	11.49465, 78.05912	15 KM/Hr
40	Turn	Medium	11.49458, 78.05989	30 KM/Hr
41	Turn	High	11.49451, 78.05995	15 KM/Hr
42	Turn	High	11.49309, 78.05982	15 KM/Hr
43	Turn	High	11.49303, 78.05997	15 KM/Hr
44	Turn	Medium	11.49362, 78.06091	30 KM/Hr
45	Turn	High	11.49366, 78.06136	15 KM/Hr
46	Turn	Medium	11.49378, 78.06146	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
47	Turn	Medium	11.49553, 78.06186	30 KM/Hr
48	Turn	Medium	11.49563, 78.06201	30 KM/Hr
49	Turn	Medium	11.49660, 78.06782	30 KM/Hr
50	Turn	Medium	11.49669, 78.06790	30 KM/Hr
51	Turn	High	11.49730, 78.06790	15 KM/Hr
52	Turn	High	11.49738, 78.06877	15 KM/Hr
53	Turn	High	11.49914, 78.06944	15 KM/Hr
54	Turn	Medium	11.49956, 78.07103	30 KM/Hr
55	Blind Spot	Blind Spot	11.50124, 78.07191	10 KM/Hr
56	Turn	Medium	11.50070, 78.07343	30 KM/Hr
57	Turn	Medium	11.50092, 78.07374	30 KM/Hr
58	Turn	Medium	11.50085, 78.07854	30 KM/Hr
59	Turn	Medium	11.50374, 78.08048	30 KM/Hr
60	Turn	High	11.50424, 78.08040	15 KM/Hr
61	Blind Spot	Blind Spot	11.51452, 78.08915	10 KM/Hr
62	Turn	High	11.51417, 78.09091	15 KM/Hr
63	Turn	Medium	11.51439, 78.09102	30 KM/Hr
64	Blind Spot	Blind Spot	11.51494, 78.09165	10 KM/Hr
65	Blind Spot	Blind Spot	11.51485, 78.09173	10 KM/Hr
66	Turn	High	11.51867, 78.09370	15 KM/Hr
67	Turn	Medium	11.51839, 78.09601	30 KM/Hr
68	Turn	Medium	11.52171, 78.10101	30 KM/Hr
69	Turn	Medium	11.52180, 78.10119	30 KM/Hr
70	Turn	High	11.52196, 78.10131	15 KM/Hr
71	Turn	High	11.52269, 78.10117	15 KM/Hr
72	Turn	Medium	11.52261, 78.10307	30 KM/Hr
73	Turn	Medium	11.52220, 78.10470	30 KM/Hr
74	Turn	Medium	11.52495, 78.10938	30 KM/Hr
75	Turn	Medium	11.52886, 78.11584	30 KM/Hr
76	Turn	Medium	11.52944, 78.11589	30 KM/Hr
77	Turn	Medium	11.52959, 78.11601	30 KM/Hr
78	Turn	Medium	11.52950, 78.11777	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
3	hospital	Government Hospital, Vennandur	11.51701, 78.088631	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	college	Mahendra Engineering College	11.4770412, 78.0024214	30 km/h	Medium
1	school	Akv Schools	11.4850407, 78.0227236	30 km/h	Medium
2	school	Akv Schools	11.4854328, 78.0233558	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.44861, 77.87429



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45478, 77.88925



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.43675, 77.91693



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.43684, 77.91706



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.43926, 77.91705



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.43930, 77.92282



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.43967, 77.92345



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.43979, 77.93009



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44030, 77.93066



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44067, 77.93203



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44088, 77.93223



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44127, 77.93218



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44163, 77.93244



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44243, 77.93244



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44285, 77.93440



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44093, 77.93821



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44550, 77.95329



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44369, 77.96036



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44391, 77.96277



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49063, 78.02680



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49135, 78.03252



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49183, 78.03257



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49315, 78.03664



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49355, 78.03703



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49374, 78.04232

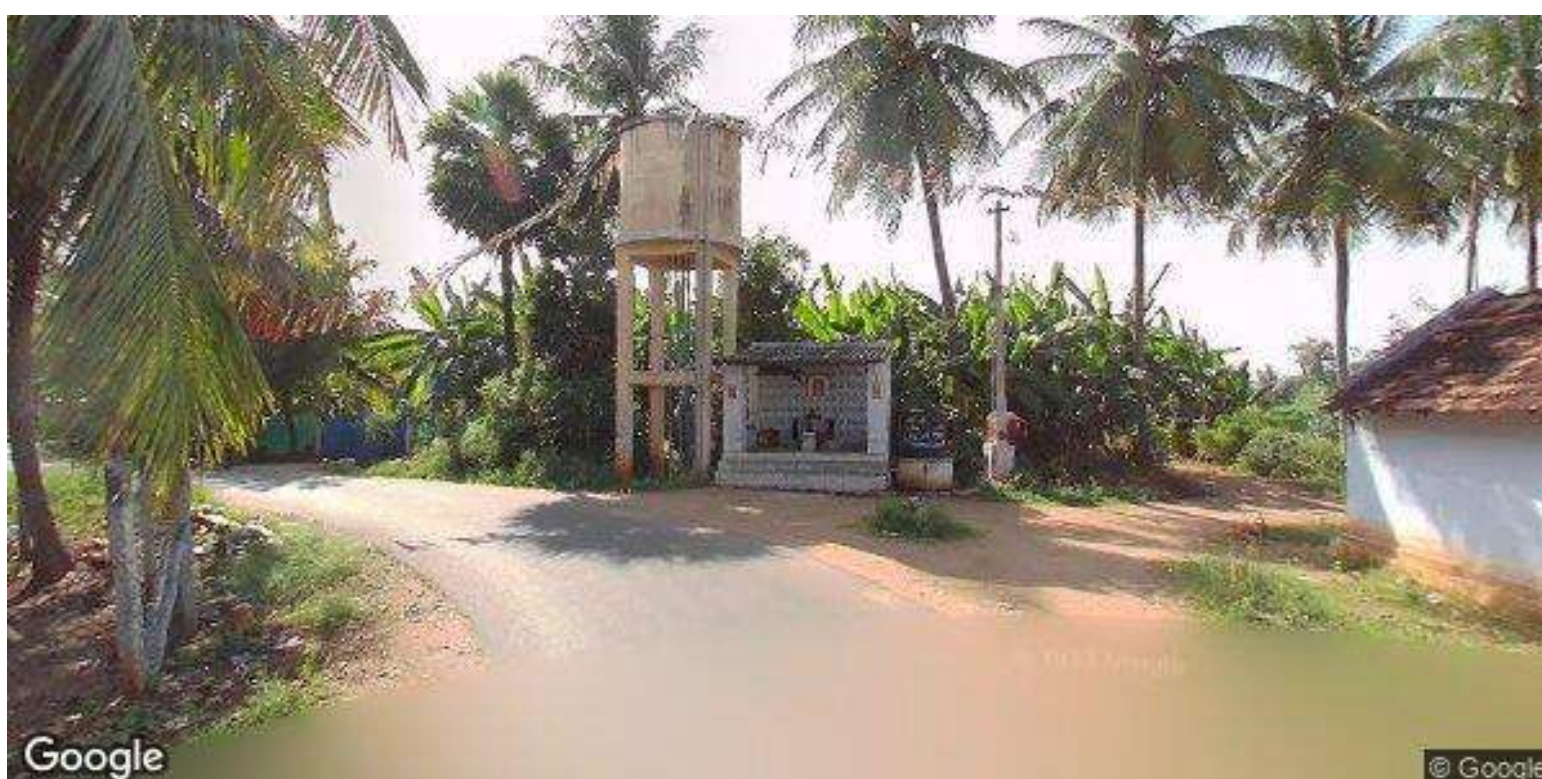


Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49645, 78.04990



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49676, 78.05010



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49723, 78.05308



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49553, 78.05338



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49547, 78.05442



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49518, 78.05498



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49544, 78.05538



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49525, 78.05686



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49546, 78.05719



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49548, 78.05904



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49477, 78.05905



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49465, 78.05912



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49458, 78.05989



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49451, 78.05995



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49309, 78.05982



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49303, 78.05997



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49362, 78.06091



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49366, 78.06136



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49378, 78.06146



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49553, 78.06186



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49563, 78.06201



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49660, 78.06782



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49669, 78.06790



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49730, 78.06790



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49738, 78.06877



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49914, 78.06944



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49956, 78.07103



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.50124, 78.07191



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50070, 78.07343



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50092, 78.07374



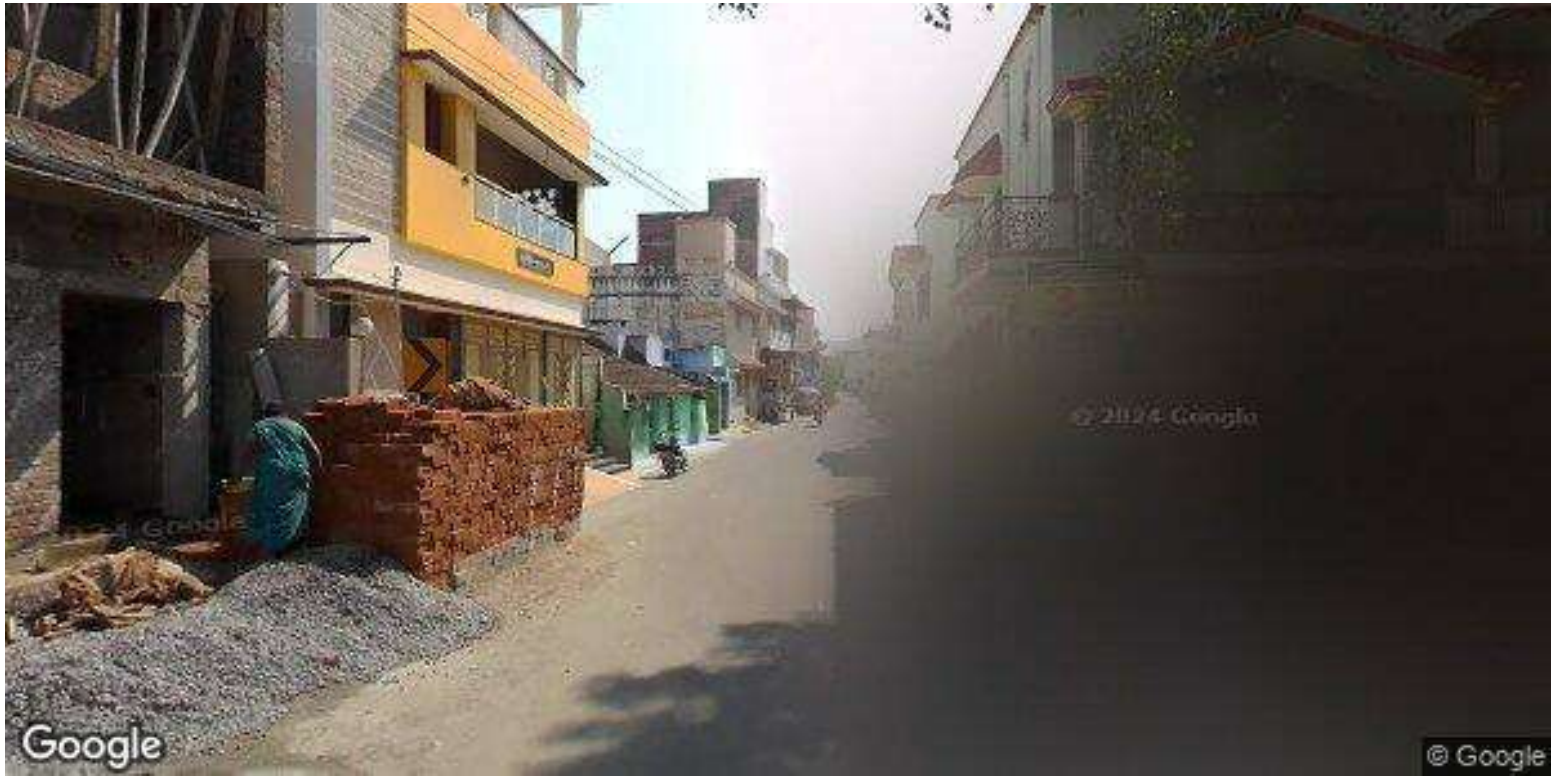
Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50085, 78.07854



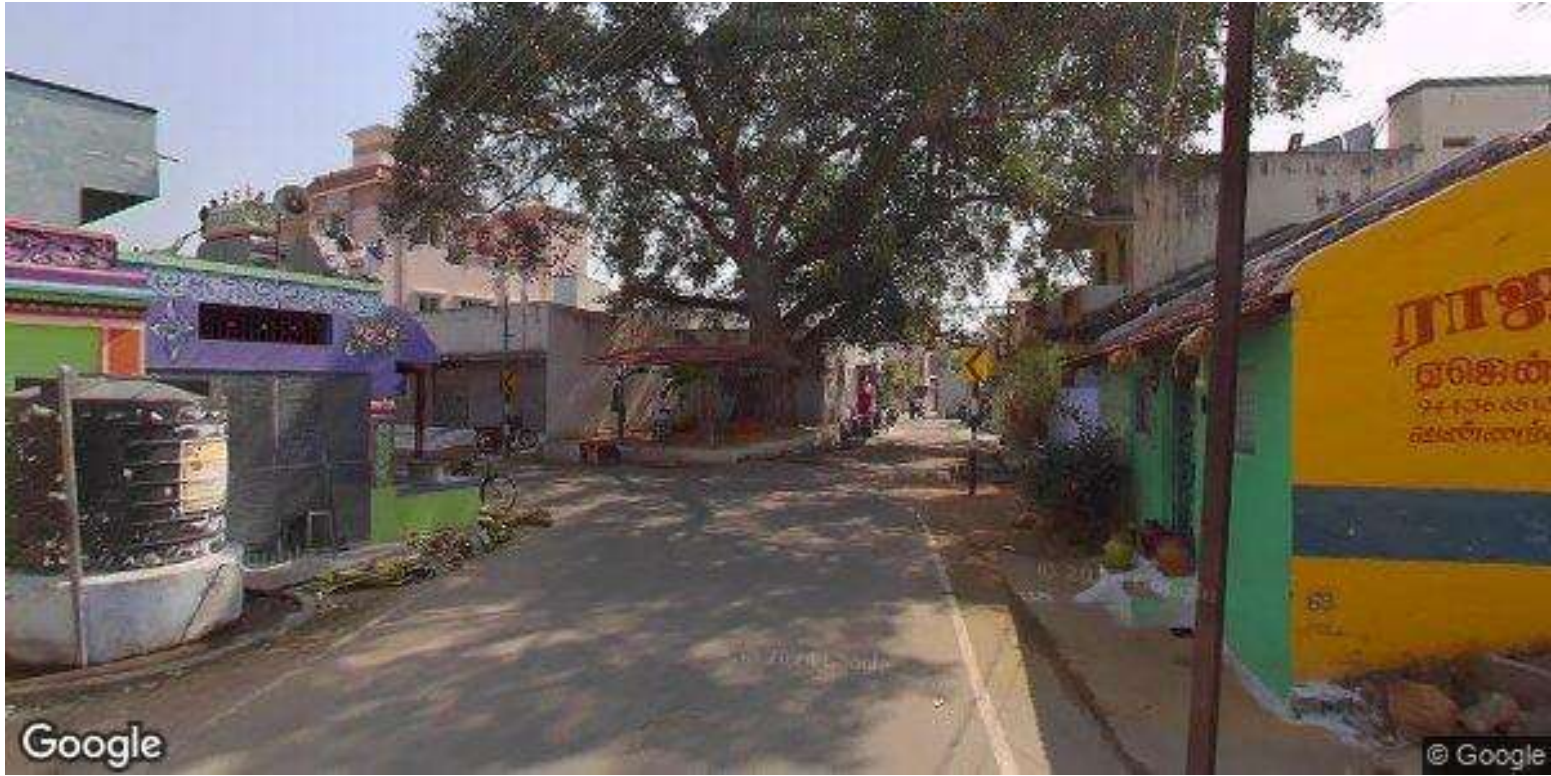
Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50374, 78.08048



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.50424, 78.08040



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.51452, 78.08915



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.51417, 78.09091



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.51439, 78.09102



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.51494, 78.09165



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.51485, 78.09173



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.51867, 78.09370



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.51839, 78.09601



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.52171, 78.10101



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.52180, 78.10119



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.52196, 78.10131



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.52269, 78.10117



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.52261, 78.10307



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.52220, 78.10470



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.52495, 78.10938



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.52886, 78.11584



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.52944, 78.11589



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.52959, 78.11601



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.52950, 78.11777